

DFR0654-F supports WiFi and Bluetooth dual-mode communication and features a small size, ultra-low power consumption, onboard charging circuit, and easy-to-use interface, which can be conveniently used for smart home IoT, industrial IoT applications, wearable devices, and so on. You can easily create your own IoT smart home system when connecting it with an IoT platform like IFTTT.

FireBeetle 2 ESP32-E supports [Arduino](#) programming and will support Scratch graphical programming and MicroPython programming very soon. We provide you with detailed online tutorials and application cases, and there are thousands of sensors with welding-free Gravity interfaces and actuators to help you get started easily. Besides, the stamp hole design makes it able to be easily embedded in your PCB, greatly saving your costs and time to build and test a prototype.

We sell two versions of Firebeetle 2 ESP32-E, unsoldered version and with header version. If you want to plug the Firebeetle shield into your Firebeetle mainboard, the Firebeetle 2 Board ESP32-E(Pre-soldered) is better for you.

Note: FireBeetle 2 and FireBeetle pins and sizes are not compatible, and neither are the expansion boards.

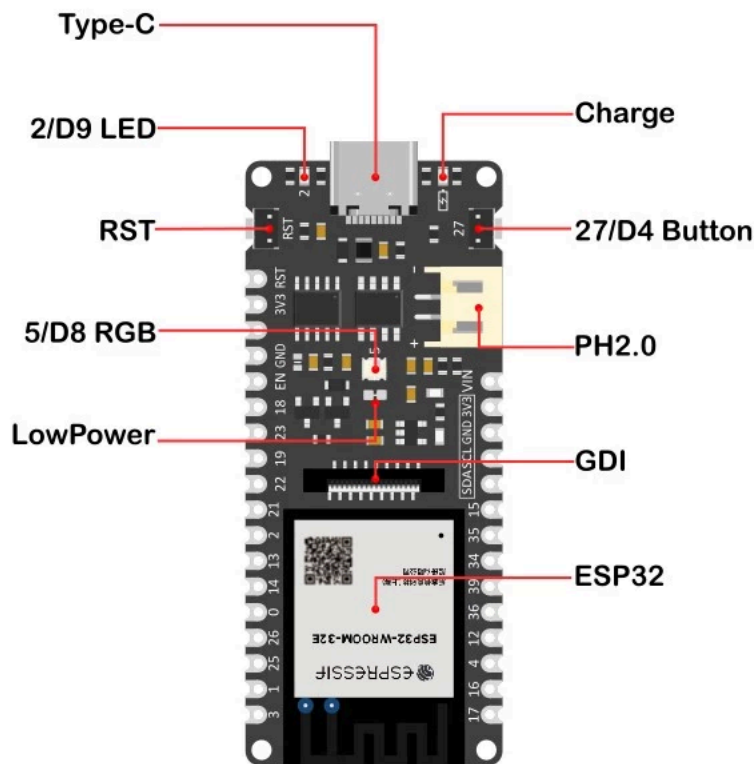


Figure 1. Board Overview

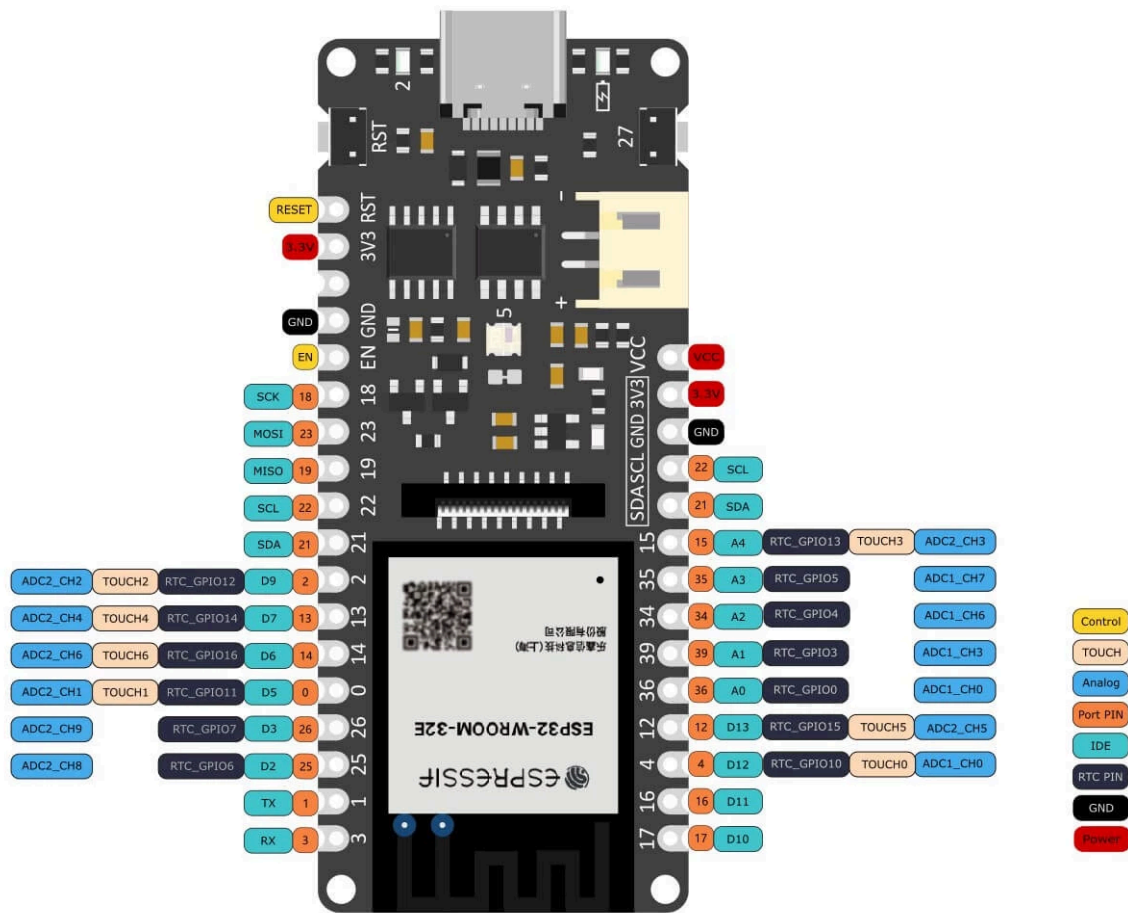


Figure 2.Pinout

Features

- ESP32 dual-core low-power main controller with power consumption as low as 2mA
- WiFi+Bluetooth 4.0 Dual-mode Module
- GDI Display Port, easy to connect
- Onboard Charging Circuit and PH2.0 lithium Battery Interface

Specification

Power Parameters

- Input Voltage
- USB-C Port: 5V DC
- PH2.0 Connector: 3.7~4.2V Li-ion
- VCC Pin: 5V DC

MCU Parameters

Processor: Xtensa Dual-Core 32-bit LX6 Microprocessor

Clock Speed: 240 MHz

SRAM: 520 KB

ROM: 448 KB

Flash Memory: 4 MB

Wireless Parameters - Wi-Fi Protocol:

IEEE 802.11 b/g/n (Data rate up to 150 Mbps in 802.11n mode)

Supports A-MPDU and A-MSDU aggregation

Supports 0.4 μ s guard interval

Operating Channel Center Frequency Range: 2412~2484 MHz

Bluetooth Protocol:

Bluetooth V4.2 BR/EDR and Bluetooth LE standards

Bluetooth CVSD and SBC audio

Bluetooth Frequency Range: 2402~2480 MHz

Peripheral Parameters

Digital Pins $\times$ 18: IO0, IO1, IO2, IO3, IO4, IO12, IO13, IO14, IO15, IO16, IO17, IO18, IO19, IO21, IO22, IO23, IO25, IO26

Analog Pins $\times$ 11: IO0, IO2, IO4, IO12, IO13, IO14, IO15, IO25, IO26, I34, I35

UART Interfaces: $\times$ 2

SPI Interface: $\times$ 1

I2C Interface: $\times$ 1

I2S Interface: $\times$ 1

DAC Interfaces: $\times$ 2

Touch Interfaces: $\times$ 7

LED PWM Channels: $\times$ 16

RGB_LED: WS2812

Display Interface: GDI

 **FireBeetle 2 ESP32-E IoT Microcontroller with Header (Supports Wi-Fi & Bluetooth)** **\$9.90**

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Module Size: 25.4mm $\times$ 60mm

Mounting hole size: hole diameter 2.0mm

Weight: 23g


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